



Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and product preferences across **3,900 transactions**

Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

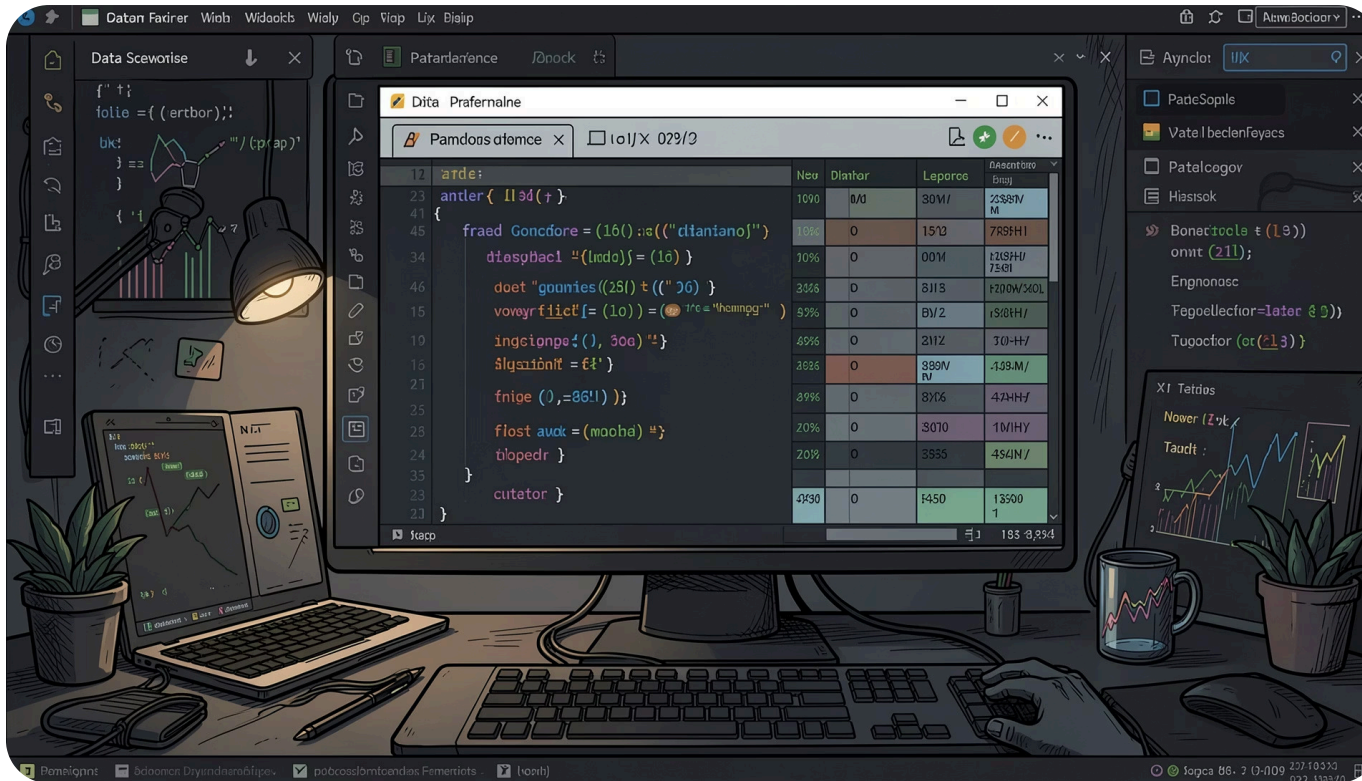
50

Locations

Key features: demographics, purchase details, shopping behavior, subscription status



Data Preparation in Python



→ Clean & Load

Imported via pandas, checked structure with `df.info()`

→ Impute Missing Data

Median rating per category for Review Rating

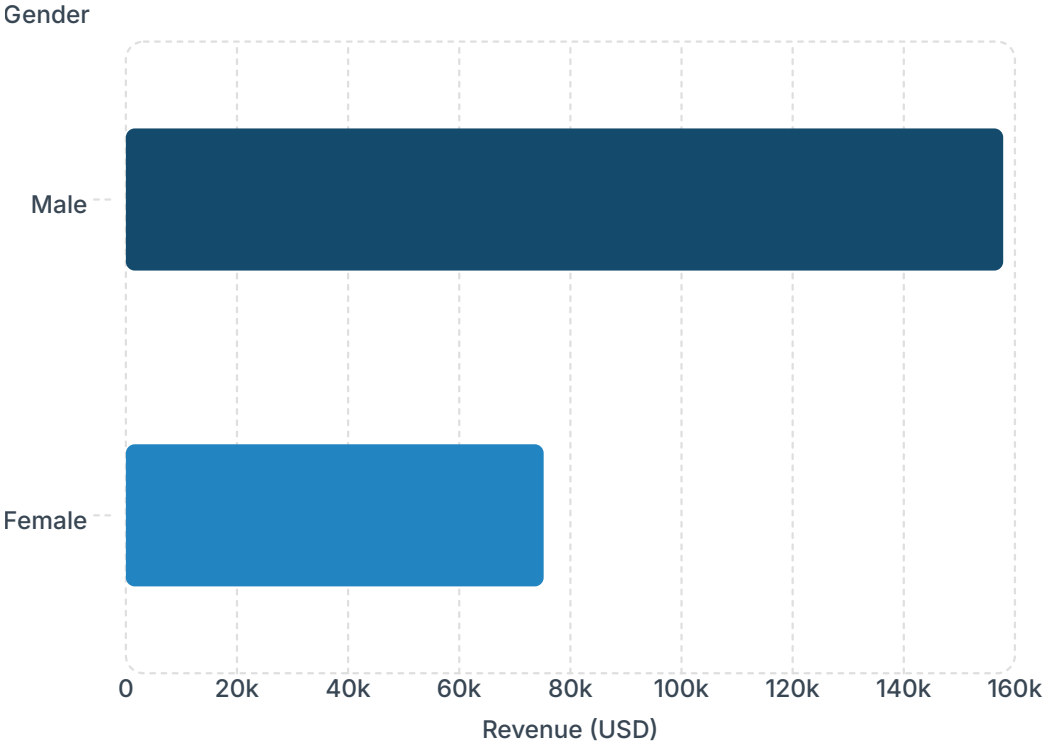
→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ PostgreSQL Integration

Loaded cleaned DataFrame for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2× more revenue** than female customers — a significant gap worth investigating further.

Male

\$157,890

Female

\$75,191

Product & Shipping Insights

Top 5 Products by Rating

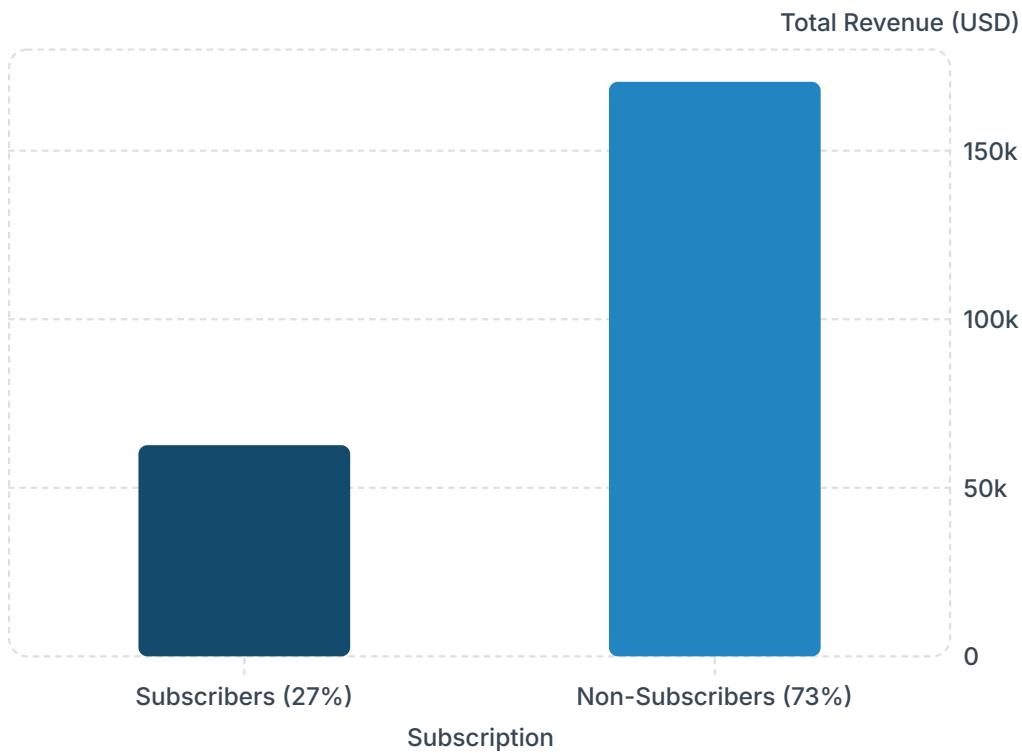
Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type – Avg Spend

Type	Avg (\$)
Standard	\$58.46
Express	\$60.48

 Express shippers spend **\$2 more** on average

Subscribers vs. Non-Subscribers



What the Data Shows

Non-subscribers drive **73% of customers** and majority of revenue, but subscribers have similar average spend.

1,053 Subscribers

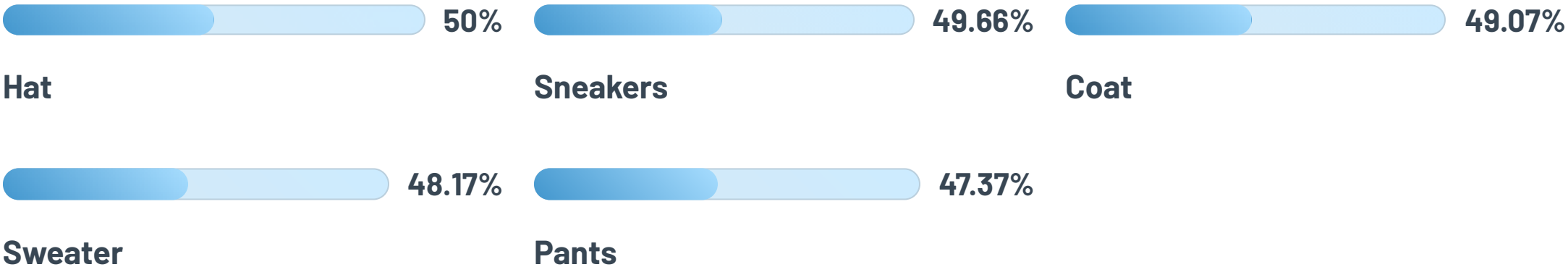
Avg spend: \$59.49

2,847 Non-Subscribers

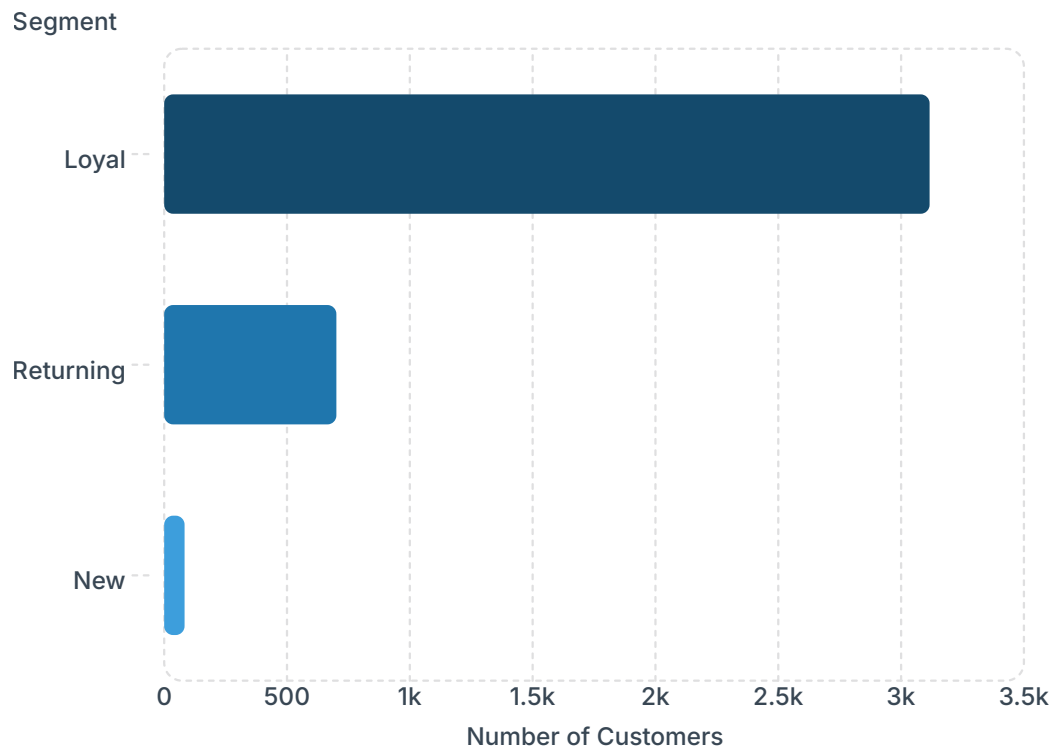
Avg spend: \$59.87

Discount-Dependent Products

Products most frequently purchased with a discount applied



Customer Segmentation



Segment Breakdown



Loyal

3,116 customers — **80%**
of base



Returning

701 customers — **18%**



New

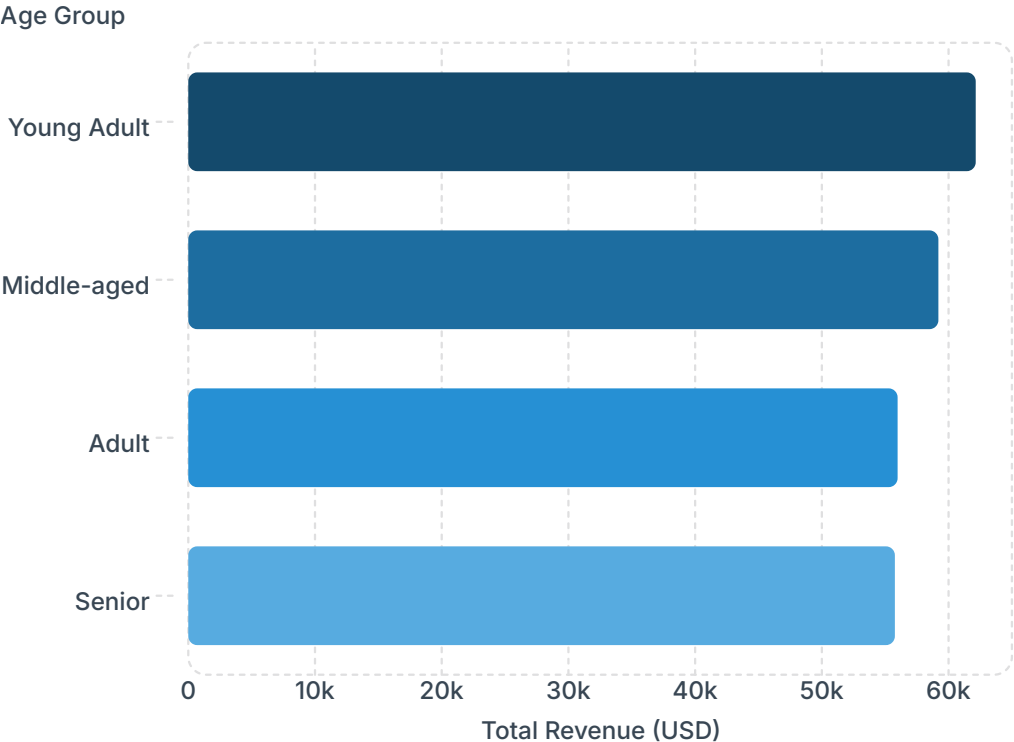
83 customers — **2%**



Of 958 repeat buyers (>5 purchases), most are **not**
subscribed

Revenue by Age Group & Category

Revenue by Age Group



Top Products per Category

Category	Top Product
Accessories	Jewelry (171 orders)
Clothing	Blouse (171 orders)
Footwear	Sandals (160 orders)
Outerwear	Jacket (163 orders)

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discounts

Balance sales boost with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers



Product Positioning

Highlight top-rated & best-selling items